Thomas Coen

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EDUCATION

Duke University, Pratt School of Engineering **Durham, NC**
B.S. Mechanical Engineering, Class of 2029 *GPA: 3.78*
Relevant coursework: Thermodynamics, Differential Equations, Engineering Design

TECHNICAL EXPERIENCE

Duke AERO (Rocketry Team) **Durham, NC**
Structures Lead *2026 – Present*
- Responsible for all structural components of the team's two-stage 30k SRAD rocket for IREC 2027: airframe, fin can, staging interface, and recovery structure
- Growing the structures subteam from 3 members through fall recruitment

Structures & Solid Propulsion Member *2025 – 2026*
- Machined 6061 aluminum nozzle casing on CNC lathe; hardware survived full-duration static hot fire
- Cut prepreg carbon fiber and performed layup for rocket body tubes
- Machined canard and air brake housing components
- Assembled motor hardware for static hot fire testing

BreakerBots, FRC Team 5104 **Pacific Grove, CA**
Co-Captain, CAD & Manufacturing Lead, Drive Coach *2022 – 2025*
- Redesigned 3-stage cascade elevator (9.5 ft travel, 2× Kraken X60, 1:5 reduction) with constant-force spring assist sized to offset moving mass; cut measured extension time from 3.5 s to 1.8 s, enabling an estimated 25% increase in scoring cycles
- Cut robot weight from 128 lb to 114 lb between the first and competition builds by pocketing aluminum structure and replacing machined brackets with 3D-printed carbon fiber, clearing the 115 lb competition limit
- Redesigned end effector to a lighter geometry with better game piece retention and more reliable handoff from the intake funnel
- Coordinated mechanical integration across 5 subteams and 35 members through a shared build schedule
- Progressed from machinist (2022) to Manufacturing Lead (2023) to Manufacturing Lead (2024) to Co-Captain (2025)

Duke Pratt First-Year Design (EGR 101) — "Bark-itects" **Durham, NC**
Mechanical Design *2025*
- Designed a wireless alert system letting a diabetes service dog notify its handler discreetly, built for a client with Type 1 Diabetes; did all CAD for a 4-person team
- CAD'd and 3D printed both enclosures (dog-mounted sensor housing, handler bracelet); final device weighed 41 g against a 317 g requirement
- Iterated the pressure sensor and bringsel through 4 design revisions to tune the actuation threshold, eliminating both missed triggers and false positives from routine dog movement through a dog-safe biothane exterior
- Delivered a prototype validated at 91.67% transmission reliability over 100 trials; documented 6.33 hr battery life as the primary shortfall against a 9 hr requirement
- Innovation in Design Award, Duke University

Duke EGR 121 — Sunflower Seed Sheller **Durham, NC**
Mechanical Design *2026*
- Designed the rollers, drive mechanism, and funnel; did all CAD for a 4-person team
- Iterated roller geometry to crack each seed along its seam instead of crushing it at random, eliminating the kernel damage seen in early prototypes
- Produced 41 intact kernels in a 3-minute run — top score of 8 teams on a yield-minus-waste metric

SKILLS

Design: SolidWorks, Onshape, Fusion 360 (CAM + CAD), AutoCAD
Manufacturing: CNC milling and turning (Tormach 1100MX/770M), manual Z-axis milling, manual lathe, 3D printing, carbon fiber layup, filament winding, laser cutting; Titans of CNC certification series
Programming: Java, Python